

Natural classes of matrices in Euclidean space

I will summarise the main results for linear operators $T : \mathbb{C}^n \rightarrow \mathbb{C}^n$ ($T : \mathbb{R}^n \rightarrow \mathbb{R}^n$) which are related to the inner product on $\mathbb{C}^n$ ($\mathbb{R}^n$).

Often I will confuse the operator and the matrix of the operator with respect to some basis (the basis being understood from context).

If the proofs are not relatively straightforward I will indicate this.

0.1 Existence of Good Orthonormal Basis

- The operator T is *self adjoint* iff it corresponds to a (real) stretch in n orthogonal directions ($\mathbb{C}$ and $\mathbb{R}$ case).

Note that in the $\mathbb{C}$ case, a “direction” corresponds to a 1-complex-dimensional subspace and hence to a 2-real-dimensional subspace. A stretch corresponds to multiplying by a real number in each of these n complex subspaces.

Equivalently, T is *self adjoint* iff there is an orthonormal basis w.r.t. which T is a real diagonal matrix.

$$T = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}$$

Think of the unit ball at the origin being stretched into an ellipsoid.

- U is *unitary* ($\mathbb{C}$ case) iff it corresponds to a rotation (multiplication by $e^{i\theta_1}, \dots, e^{i\theta_n}$) in n orthogonal directions $v_1, \dots, v_n$.

Equivalently, U is *unitary* iff there is an orthonormal basis w.r.t. which U is a diagonal matrix with all diagonal entries having absolute value 1.

$$U = \begin{bmatrix} e^{i\theta_1} & & \\ & \ddots & \\ & & e^{i\theta_n} \end{bmatrix}$$

The corresponding result for *orthonormal* matrices ($\mathbb{R}$ case) is not so clean and it is better to *define* orthonormal matrices as we do in ***. The result is that O is orthonormal iff there is an orthonormal basis with respect to which O has matrix

$$O = \begin{bmatrix} 1 & & & & & & & \\ & \ddots & & & & & & \\ & & 1 & & & & & \\ & & & -1 & & & & \\ & & & & \ddots & & & \\ & & & & & -1 & & \\ & & & & & & \cos \theta_1 & -\sin \theta_1 \\ & & & & & & \sin \theta_1 & \cos \theta_1 \\ & & & & & & & \ddots & \\ & & & & & & & & \cos \theta_k & -\sin \theta_k \\ & & & & & & & & \sin \theta_k & \cos \theta_k \end{bmatrix}$$

Thus there is an orthonormal basis such that O fixes some vectors, reflects others, and rotates in orthogonal planes determined by couples of the remaining basis vectors. In particular, in $\mathbb{R}^3$ every orthonormal operator is a rotation around some vector followed perhaps by a reflection in the plane orthogonal to this vector!

Think of the unit ball at the origin being rotated into itself (perhaps there is also a reflection, thus reversing the orientation)

- T (in the $\mathbb{C}$ case) is *normal* iff it corresponds to multiplication by a (possibly different) complex number in each of n orthogonal directions. That is, to a stretch followed by a rotation in each of the corresponding n 2-real-dimensional subspaces.

Equivalently, iff there is an orthonormal basis with respect to which T is diagonal (with complex, not necessarily real, entries).

0.2 Adjoints

- The adjoint T^* of the unique operator T is defined by $\langle T^*u, v \rangle = \langle u, Tv \rangle$ for all $u, v \in \mathbb{C}^n$ ($\mathbb{R}^n$).
- With respect to any orthonormal basis, the matrix T^* is the conjugate (not necessary of course in the $\mathbb{R}$ case) of the transpose of T , i.e. $T_{ij}^* = \overline{T_{ji}}$.

0.3 Analytic and Matrix Approaches

- T is self adjoint iff $T = T^*$.

Because of what this means for matrices, one often says T is *Hermitian* in the $\mathbb{C}$ case and *symmetric* in the $\mathbb{R}$ case.

The $\implies$ case is easy but the converse requires work, It follows easily from the more general result for normal operators below

- U is unitary (O is orthonormal) iff $UU^* = U^*U = I$ ($OO^* = O^*O = I$). (Clear from the matrix version)

For matrices this is equivalent to the columns being orthonormal (norm one and inner products of different columns equalling zero) and also equivalent to the rows having the same property.

- N is normal iff $TT^* = T^*T$. ($\implies$ is clear from the definition. The converse requires work).
- Trivially, unitary $\implies$ normal and self adjoint $\implies$ normal.

0.4 More remarks on U and O

A matrix U is unitary (O is orthogonal) iff it preserve norms iff it preserve inner products iff it sends some orthonormal basis to another orthonormal basis iff it sends every orthonormal basis to an orthonormal basis.

That is U is unitary iff

- $\|Uv\| = \|v\|$ for every v
- $\langle Uv, Uw \rangle = \langle v, w \rangle$ for all v, w
- $v_1, \dots, v_n$ orthonormal $\implies Uv_1, \dots, Uv_n$ orthonormal

Similarly for “orthogonal”.

0.5 Polar Decomposition

- Every linear operator T ($\mathbb{C}$ case) can be written in the form $T = UH$ where U is unitary and H is Hermitian (i.e. self adjoint) with nonnegative e-values. That is, real nonnegative stretching in orthogonal directions followed by rotations in another set of orthogonal complex directions.

(If the rotations are in the same complex directions as the stretches, then the matrix is clearly normal.)

- In the $\mathbb{R}$ case $T = OS$ where O is orthonormal and S is symmetric with nonnegative e-values. That is, T is again a nonnegative stretch in orthogonal directions and then apply orthogonal transformation.

Think of taking the unit ball sitting at the origin, stretching it to an ellipsoid, and then rotating.

0.6 Another Characterisation

- T is selfadjoint iff $\langle Tu, u \rangle$ is real for all u
- T is unitary (orthonormal) iff $\|Tu\| = \|T^*u\| = \|u\|$ for all u
- T is normal iff $\|Tu\| = \|T^*u\|$ for all u

The proofs are fairly straightforward (with the HINT from the relevant assignment problem).